HOSPITAL OPERATIONS & REVENUE ANALYSIS

(Using Excel, SQL & Power BI)

Submitted By: Md Adil

[Noida Institute of Engineering and Technology]

ABSTRACT

This project analyzes hospital operations to understand patient flow, revenue trends, and treatment performance. Data was first cleaned using Microsoft Excel to ensure accuracy and consistency. SQL was used to extract and transform the data, and Power BI was used to build an interactive dashboard. The analysis helps identify key trends, inefficiencies, and opportunities for improving hospital performance.

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1. INTRODUCTION

Healthcare organizations generate large volumes of data daily. Analyzing this data helps improve operational efficiency and decision-making. This project focuses on analyzing hospital data to track appointments, revenue, treatments, and payment performance using modern data analytics tools.

2. OBJECTIVES

- Analyze patient demographics and behavior
- Track appointment and revenue trends
- Evaluate doctor performance
- Identify top treatments contributing to revenue
- Analyze payment methods and failures
- Build an interactive dashboard for visualization

3. DATASET DESCRIPTION

The dataset includes the following tables:

- Patients → Patient ID, Name, Gender
- Doctors → Doctor ID, Doctor Name
- Appointments → Appointment ID, Date,

  Patient ID, Doctor ID
- Treatments → Treatment ID, Treatment Type
- Billing → Billing ID, Amount, Payment Type,

  Payment Status.

4. TOOLS & TECHNOLOGIES

✓Microsoft Excel → Data Cleaning

✓SQL (PostgreSQL) → Data Analysis

✓Power BI → Data Visualization

5. DATA CLEANING (EXCEL)

Data cleaning was performed using Microsoft Excel to ensure data quality before analysis.

- Steps performed:

- Removed duplicate records
- Handled missing values
- Standardized column names
- Corrected inconsistent data formats
- Converted date columns into proper format
- Ensured uniform naming for categorical data (e.g., Gender, Payment Type)

- Outcome:

- Clean and structured dataset ready for SQL analysis
- Improved accuracy of insights and dashboard visuals

6. SQL QUERIES

* Basic Queries

* Joins (Inner Join, Left Join)

* Aggregations (SUM, COUNT, AVG)

* Subqueries

* Window Functions

```
select * from doctors;

select * from appointments;

select * from patient;

select * from billing;

select * from treatments;


--count total number of patients--

select count(*)

from patient;


--count patient by gender--

select gender,count(*) total

from patient

group by gender;


--showing patient name,doctor name,appointment date--

select p.first_name||' '||p.last_name patients_name,str_agg(d.first_name||' '||d.last_name
',' ),a.appointment_date

from patient p

inner join appointments a

on p.patient_id=a.patient_id

inner join doctors d

on d.doctor_id=a.doctor_id
```

```
group by patients_name;


--total appointments--

select count(*)

from appointments


--count number of appointment per doctor--

select d.first_name||' '||d.last_name as doctors_name,count(a.appointment_id)
num_of_appointment

from doctors d

inner join appointments a

on d.doctor_id=a.doctor_id

group by doctors_name;


--list all treatment taken by each patient--

select p.first_name||' '||p.last_name as patient_name,string_agg(t.treatment_type,' ,')
total_treatment

from patient p

inner join billing b

on p.patient_id=b.patient_id

inner join treatments t

on t.treatment_id=b.treatment_id

group by patient_name;


--average billing amount--

select avg(amount)

from billing


--show patient name,treatment,cost--

select p.first_name||' '||p.last_name as patient_name,string_agg(t.treatment_type,' ,')
total_treatment,sum(t.cost) as total_cost

from patient p
```

```
inner join billing b
on p.patient_id=b.patient_id
inner join treatments t
on t.treatment_id=b.treatment_id
group by patient_name
order by total_cost desc;

--show most common treatment--
select treatment_type,count(*) number
from treatments t
group by treatment_type
order by number desc limit 1

--total_revenue generated by hospital--
select sum(cost) total_revenue
from treatments;

--revenue generated per treatment--
select treatment_type,sum(cost)
from treatments
group by treatment_type

--revenue generated per doctor--
select d.first_name||' '||d.last_name as doctors_name, sum(t.cost) as total_revenue
from doctors d
inner join appointments a
on d.doctor_id=a.doctor_id
inner join treatments t
on t.appointment_id=a.appointment_id
group by doctors_name
order by total_revenue asc
```

```
--find monthly appointment trend--
select to_char(appointment_date,'month') as month,extract (month from appointment_date) as month_num,
count(appointment_id) num_of_appointments
from appointments
group by month,month_num
order by month_num;

--top three treatments by revenue--
select treatment_type,sum(cost) revenue
from treatments
group by treatment_type
order by revenue desc limit 3

--patient who visited more than ones--
select p.first_name||' '||p.last_name as patient_name,count(a.appointment_id) as num
from patient p
inner join appointments a
on p.patient_id=a.patient_id
group by patient_name
having  count(a.appointment_id)>1;

--ranking doctors based on revenue--
select d.first_name||' '||d.last_name as doctors_name,sum(t.cost) as revenue,rank() over(order by sum(t.cost) desc) as rank
from doctors d
 inner join appointments a
on d.doctor_id=a.doctor_id
inner join treatments t
on t.appointment_id=a.appointment_id
```

```
group by doctors_name;

--find buisest day at hospital--
select appointment_date,count(appointment_id) as appointment
from appointments
group by appointment_date
order by appointment desc limit 1;

--find which doctors handle most appointments--
select d.first_name||' '||d.last_name as doctors_name,count(a.appointment_id) as num_of_appointment
from doctors d
inner join appointments a
on d.doctor_id=a.doctor_id
group by doctors_name;

--find av billing amount per patient--
select p.first_name||' '||p.last_name as patient_name,avg(t.cost)
from treatments t
inner join appointments a
on t.appointment_id=a.appointment_id
inner join patient p
on p.patient_id=a.patient_id
group by p.patient_id;

--total doctors in hospital--
select count(*) as num_of_doctors
from doctors
```

```
--finding highest revenue day--
select bill_date,sum(amount) as total_amount
from billing
group by bill_date
order by sum(amount) desc limit 1

--retriveing patient details with above average billing--
select *
from patient
where patient_id in(select patient_id
                    from billing
                    where amount >(select avg(amount)
                    from billing))

--doctors with above average appointments--
select first_name||' '||last_name as doctors_name
from doctors
where doctor_id in(select doctor_id
                   from appointments
                   group by doctor_id
                   having count(*)>(select avg(cnt)
                   from (select count(*) as cnt
                   from appointments
                   group by doctor_id)))

select * from doctors;
select * from appointments;
select * from patient;
select * from billing;
select * from treatments;
```

```
--patients per gender percentage--
select gender,count(*)*100/(select count(*) from patient) as gender_percent
from patient
group by gender;

--retrive name of oldest patient--
select first_name||' '||last_name as patient_name
from patient
where date_of_birth=(select min(date_of_birth)
                                    from patient
                                    )
--retrive name of youngest patient--
select first_name||' '||last_name as patient_name
from patient
where date_of_birth=(select max(date_of_birth)
                                    from patient
                                    );

        select inet_server_addr(),ine

--monthly revenue trend--
SELECT
  TO_CHAR(b.bill_date, 'Mon') AS month,extract (month from b.bill_date) as month_num,
  SUM(b.amount) AS total_revenue
FROM billing b
GROUP BY
  TO_CHAR(b.bill_date, 'Mon'),
  EXTRACT(YEAR FROM b.bill_date),
  EXTRACT(MONTH FROM b.bill_date)
ORDER BY
  EXTRACT(YEAR FROM b.bill_date),
```

```
    EXTRACT(MONTH FROM b.bill_date)


--payment method--
select payment_method,count(*)
from billing
group by payment_method


--appointment date--
select distinct ap(appointment_date,'dd-mon-yyyy')
from appointments
order by appointment_date asc
--payment status--
select payment_method,payment_status,count(*)
from billing
group by payment_method,payment_status
```

7. DATA MODELING

A star schema model was used in Power BI:

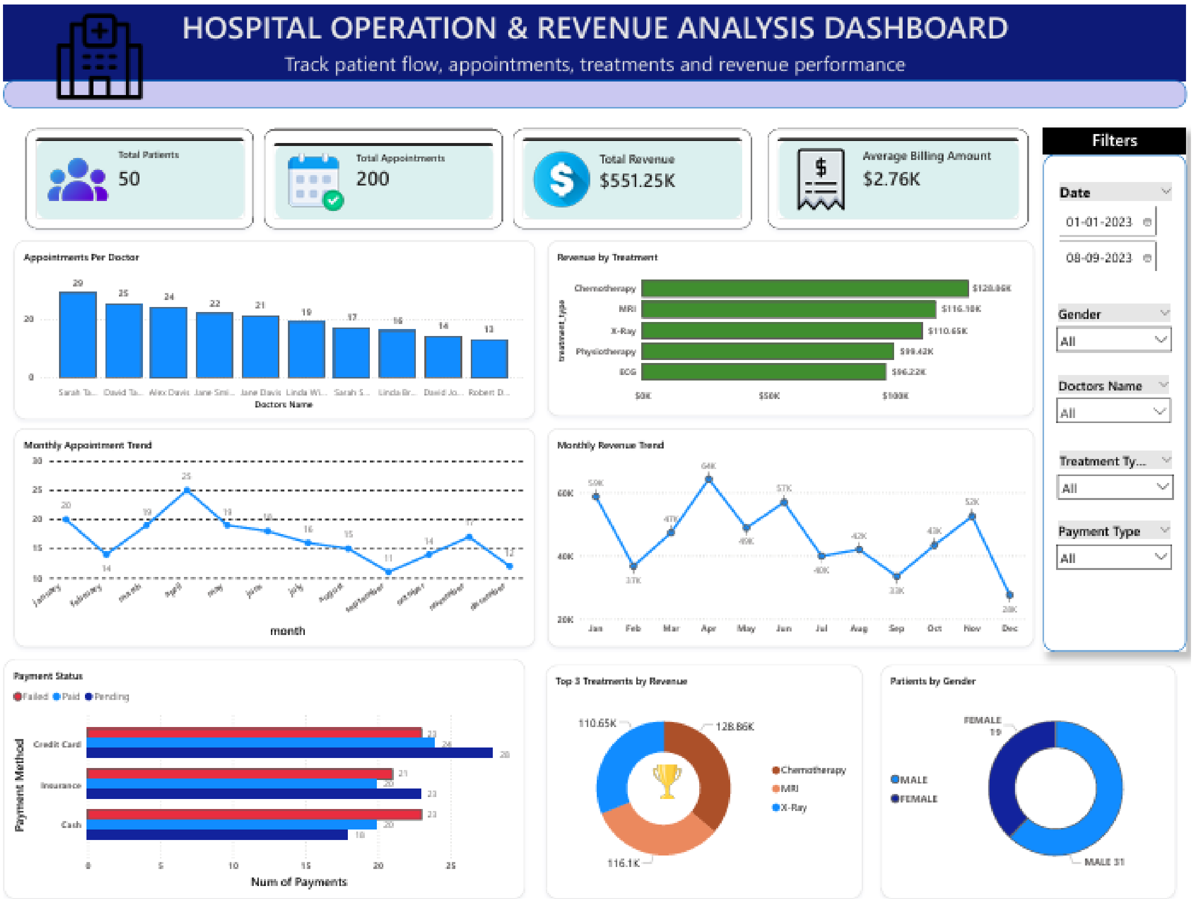
* Central table: Appointments (Fact Table)
* Connected tables:
    * Patients
    * Doctors
    * Treatments
    * Billing

Key Relationships:

* Patients → Appointments
* Doctors → Appointments
* Treatments → Appointments
* Appointments → Billing

This structure ensures proper filtering and interaction across visuals.

8. DASHBOARD OVERVIEW



The dashboard includes:

* KPI Cards:

- Total Appointments
- Total Revenue
- Average Billing

* Charts:

- Appointments per Doctor

- Revenue by Treatment
- Monthly Appointment Trend
- Monthly Revenue Trend
- Payment Analysis

* Filters:

- Date Range
- Gender
- Doctor Name
- Treatment Type
- Payment Type

9. INSIGHTS & ANALYSIS

➢ Overall Performance

* The hospital handled 200 appointments with 50 patients and generated $551K revenue

* Average billing amount is $2.76K

➢ Patient Demographics

* Male patients: 31

* Female patients: 19

- Indicates higher male patient ratio

➢ Doctor Performance

* Top doctor handled 29 appointments

* Uneven workload distribution among doctors

➢ Treatment Analysis

* Top treatments:

- Chemotherapy
- MRI

- X-Ray

* Revenue is concentrated on a few treatments

➢Appointment Trend

* Peak in March

* Lowest in February

* Indicates variation in patient visits

➢Revenue Trend

* Revenue follows appointment pattern

* Higher appointments = higher revenue

➢Payment Analysis

* Payment methods: Cash, Insurance, Credit Card

* Presence of failed and pending payments

* Indicates revenue delays and inefficiencies

10. BUSINESS RECOMMENDATIONS

* Improve payment processing system

* Reduce failed and pending payments

* Distribute workload evenly among doctors

* Promote more treatment options(Because recession in any one top treatment can affect large revenue source of hospital)

* Analyze low-performing months for improvement

  - It can be due to seasonal changes

11. CONCLUSION

The analysis shows that the hospital has strong revenue generation and steady patient flow. However, improvements can be made in payment systems, workload distribution, and service diversification. The dashboard provides a comprehensive view of hospital operations and supports data-driven decision-making.